The University of Sussex
SCHOOL OF ENGINEERING & INFORMATICS
Cover sheet for 876H1Z MSc Project
Interim report

CANDIDATE NUMBER:299106	**DEPARTMENT: ENGINEERING**
TITLE OF MODULE: **876H1Z MSc PROJECT**	
TITLE OF ASSIGNMENT: **INTERIM REPORT**	
Date due:	**Date submitted:**

DECLARATION:
In making this submission I declare that my work contains no examples of misconduct, such as plagiarism, collusion, or fabrication of results.

I confirm that I have talked with my project supervisor about whether ethical review will be required, and that the outcome of the discussion is included in my interim report.

Should ethical review be required, I confirm that I will submit an application before the end of the autumn term. Furthermore, if the ethical implications of my project change, I confirm that I will alert my supervisor immediately.

SIGNATURE:

- Work submitted late will be subject to the penalties set out at
- http://www.sussex.ac.uk/academicoffice/1-3-2.html
- Marks are provisional and subject to ratification by the relevant Examination Board
- Appeals can only be lodged against alleged procedural improprieties AFTER the ratification of marks
- Comments made by the tutor are intended to advise the student in developing their knowledge and abilities within the subject

NB. Tutors are normally required to return marks and feedback to students within 3 weeks of the date set for submission.

NB. Some assignments will be kept for ratification by the Exam Board. Students are encouraged to take a copy of their work before submission as it may not be returned.

Project Objectives

A Comparative Study of Deep Learning Models for Multi-task Human Activity Recognition

The objective of this project is to design, implement, and compare multiple deep learning models—such as LSTM, Transformer, CNN-LSTM, and LSTM with Attention—for multi-task human activity recognition based on sequential sensor data. The key goals include:

Developing a unified framework for modeling time-series signals;

Implementing multi-task learning architectures that simultaneously classify actions and assess status;

Conducting comprehensive performance evaluation across models using consistent metrics;

Identifying the most effective architecture for the given task;

Providing insights into model selection and potential improvements for real-world deployment.

Research Background

With the rapid advancement of wearable sensing technologies, human activity recognition (HAR) has become a cornerstone of applications in healthcare monitoring, smart environments, sports analytics, and human-robot interaction. These systems typically rely on inertial measurement units (IMUs)—such as accelerometers and gyroscopes—embedded in smartphones or wearable devices to capture time-series motion data. The core challenge lies in accurately classifying human behaviors (e.g., walking, sitting, falling) from these sequential sensor signals.

Traditional HAR approaches often depend on handcrafted feature extraction (e.g., mean, variance, frequency components) combined with shallow classifiers like Support Vector Machines (SVMs) or Decision Trees. While effective in constrained environments, these methods are limited by their reliance on domain expertise and poor generalization across users and scenarios.

In recent years, deep learning has revolutionized HAR by enabling end-to-end learning directly from raw sensor data. Various neural network architectures have been proposed, each with distinct strengths:

Convolutional Neural Networks (CNNs) excel at extracting local spatial and temporal patterns;

Long Short-Term Memory networks (LSTMs) are well-suited for modeling long-term temporal dependencies in sequential data;

CNN-LSTM hybrids combine the feature extraction power of CNNs with the sequence modeling capability of LSTMs;

Transformer-based models, leveraging self-attention mechanisms, have shown superior performance in capturing global context and long-range dependencies.

Despite the proliferation of these models, there is a lack of systematic comparison under consistent experimental settings—same dataset, preprocessing pipeline, input format, and evaluation metrics. Moreover, while multi-task learning (e.g., recognizing both action type and user state) is increasingly relevant for real-world deployment, few studies evaluate how different architectures perform in such scenarios.

Therefore, a comprehensive benchmarking study that implements, trains, and compares representative deep learning models—including LSTM, Transformer, CNN-LSTM, and attention-enhanced variants—is essential to understand their relative performance, computational efficiency, and suitability for practical HAR applications. This work aims to fill this gap by conducting an empirical comparison to guide model selection and inspire future architectural improvements in the field of human activity recognition.

Progress to Date

To date, significant progress has been made in building a data-driven framework for gait phase and pathology state recognition using lower-limb electromyographic (EMG) signals. The first stage of the project focused on data preprocessing and model development. Specifically, raw EMG signals collected from key lower-limb muscle groups (e.g., quadriceps, hamstrings, tibialis anterior, and gastrocnemius) were synchronized with corresponding gait cycle annotations and clinical labels indicating pathological conditions (e.g., post-stroke hemiparesis). To ensure consistency across subjects and improve model convergence, the EMG data were normalized using min-max scaling to bring all signal amplitudes into a uniform range [0, 1]. Subsequently, the preprocessed time-series data were segmented into fixed-length windows and converted into tensor format compatible with deep learning frameworks.

In the second phase, a series of deep neural network models were implemented using PyTorch, a widely adopted open-source machine learning library. The architectures include:

Long Short-Term Memory (LSTM) networks to capture temporal dynamics in EMG sequences;

Transformer-based models leveraging self-attention mechanisms for global context modeling;

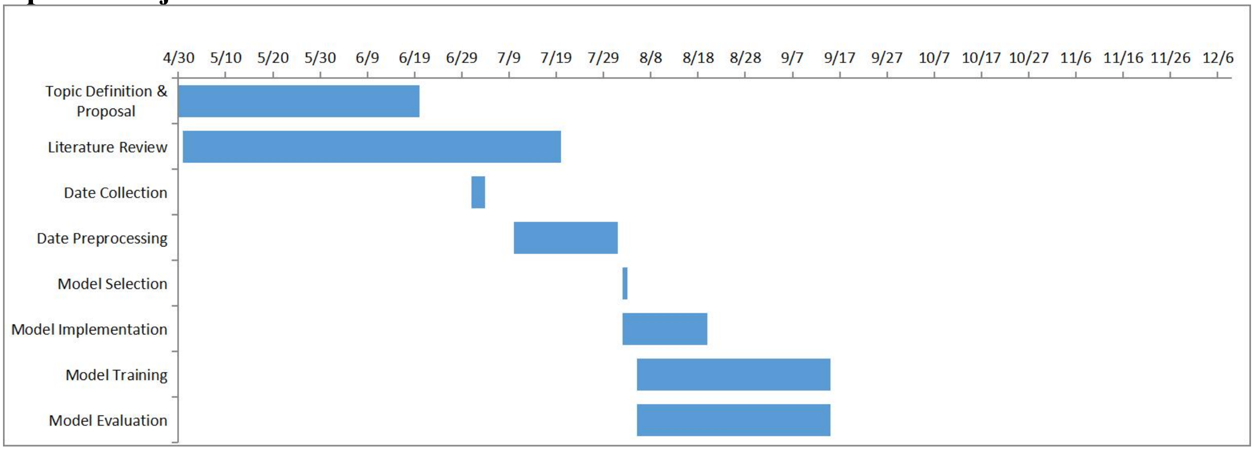
CNN-LSTM hybrid networks, where 1D convolutional layers extract local spatiotemporal features, followed by LSTM layers to model long-term dependencies;

Attention-augmented LSTM models, which apply temporal attention mechanisms to dynamically weight informative time steps in the sequence.

Each model was trained for 10,000 epochs using the Adam optimizer with a fixed learning rate of 0.001 and a batch size of 32. During training, the categorical cross-entropy loss was minimized for gait phase classification, while binary cross-entropy loss was used for pathology state detection in a multi-task learning setup. To prevent overfitting, dropout regularization (rate = 0.3) and early stopping were applied based on validation performance.

After training, the models were evaluated on a held-out test set, and their performance was quantified in terms of prediction accuracy for both gait phase and pathological state classification. The results provide a preliminary comparison of the effectiveness of different deep learning architectures in decoding human motor intent from EMG signals, laying the foundation for further model optimization and comparative analysis.

Update Project Plan



Building upon the current implementation and preliminary evaluation of deep learning models for EMG-based human activity recognition, several key directions will be pursued to further enhance the system's performance, interpretability, and practical applicability.
A systematic hyperparameter search will be conducted using techniques such as grid search, random search, or Bayesian optimization. Key parameters—including learning rate, batch size, number of layers, hidden units, and dropout rates—will be tuned to improve model convergence and generalization. This will ensure a fair and optimized comparison across architectures.
To evaluate model robustness, cross-subject validation will be performed—training on data from multiple subjects and testing on unseen individuals. Domain adaptation or few-shot learning strategies may be explored to reduce inter-subject variability and improve generalization.

A comprehensive research paper will be prepared for submission to a peer-reviewed conference or journal. The paper will present the comparative analysis, key findings, limitations, and implications for future wearable rehabilitation systems.

The writing of the research paper will follow a structured and iterative process over the coming weeks. First, a comprehensive outline will be developed, defining the key sections including Introduction, Related Work, Methodology, Experiments, Results, Discussion, and Conclusion. Based on this outline, each section will be drafted sequentially, starting with the Methodology and Results to leverage the completed implementation and experimental data. Figures and tables summarizing model performance, training curves, and attention visualizations will be integrated to enhance clarity. The draft will then undergo multiple rounds of revision for technical accuracy, logical flow, and language quality, with feedback incorporated from the advisor. Finally, the paper will be formatted according to the target journal or conference guidelines and prepared for submission, ensuring all components—including abstract, keywords, references, and captions—are complete and consistent.

Planned Work for the Rest of the Project

The remaining phase of the project will focus on model optimization, in-depth evaluation, result interpretation, and academic dissemination. Building upon the preliminary training of deep learning models (LSTM, Transformer, CNN-LSTM, and Attention-LSTM), the next steps will emphasize performance refinement, comparative analysis, and the generation of meaningful insights for human activity recognition (HAR) using lower-limb EMG signals.

The project will be implemented using a combination of widely adopted scientific computing and deep learning tools in Python. The primary deep learning framework is PyTorch, chosen for its flexibility, dynamic computation graph, and strong support for custom model development. Data preprocessing and analysis will be conducted using NumPy, Pandas, and SciPy for signal filtering, normalization, and segmentation, while scikit-learn will be used for data splitting and performance evaluation. Model training and validation will be monitored using TensorBoard to visualize loss and accuracy trends. For result visualization, Matplotlib, Seaborn, and Plotly will be employed to generate high-quality 2D plots, heatmaps, and interactive graphs. Code version control and collaboration will be managed through Git and GitHub, ensuring reproducibility and transparency. Configuration files in YAML format will be used to track hyperparameters and experimental settings across different model runs.

The experimental design aims to ensure a fair and comprehensive comparison of the deep learning models—LSTM, Transformer, CNN-LSTM, and Attention-LSTM—under consistent conditions. The input data consist of preprocessed lower-limb EMG signals segmented into fixed-length time windows (e.g., 100 ms) and normalized to a uniform scale. Each model will be trained using the same training, validation, and test splits, with a subject-independent strategy (e.g., leave-one-subject-out or 5-fold cross-validation) to evaluate generalization across individuals. Training will be conducted for up to 10,000 epochs using the Adam optimizer (learning rate = 0.001, batch size = 32), with early stopping applied based on validation loss to prevent overfitting. All models will be evaluated using standard classification metrics, including accuracy, precision, recall, F1-score, and AUC-ROC, enabling both per-class and overall performance analysis. This rigorous setup allows for a reliable assessment of each architecture's effectiveness in decoding gait phases and pathological states from EMG signals.

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